Project: Build panorama generation software for store shelves (Submitted by Prerana.B(prerana.bora120@gmail.com)

Problem statement:
Develop a system that stitches multiple overlapping images of a retail store into a single seamless panorama while preserving accurate shelf geometry and product placement.

Rules

- Generate one panorama per store.
- Include all visible fixture regions.
- Do not duplicate products in overlapping areas.
- Preserve straight shelf alignment.
- Minimize ghosting, warping, tearing, and visible seams.
- Keep product packaging and price tags readable.
- Ensure the panorama is suitable for downstream retail analysis.

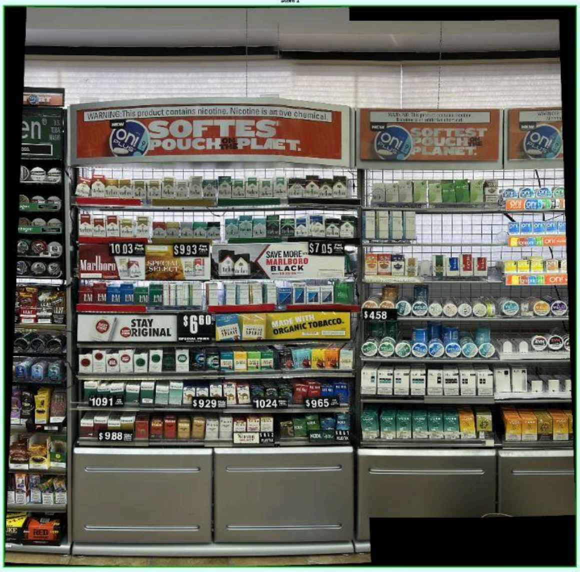
Approaches experimented:

Approach 1: OpenCV cv2.Stitcher + SIFT Fallback

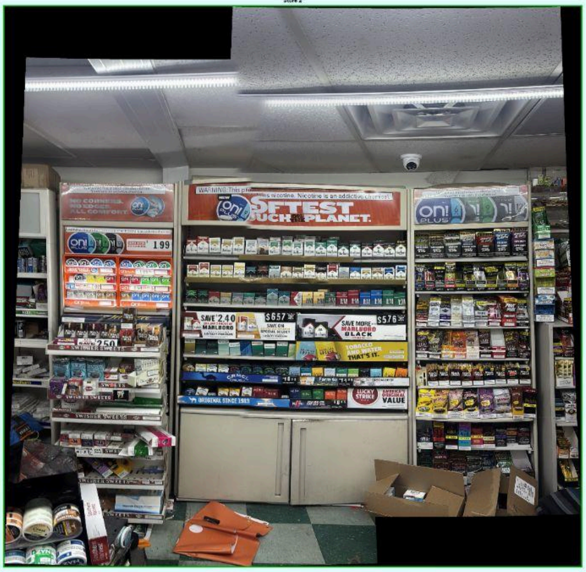
- cv2.Stitcher_PANORAMA → handles full pipeline in one call → fastest baseline
- If fails → retry at 50% resolution → stabilizes matching on high-res images
- If still fails → SIFT + FLANN + Lowe's ratio test → filters false matches from repetitive shelf patterns0.75
- RANSAC homography → rejects outliers → alpha blend → crop black borders5.0px

Results:

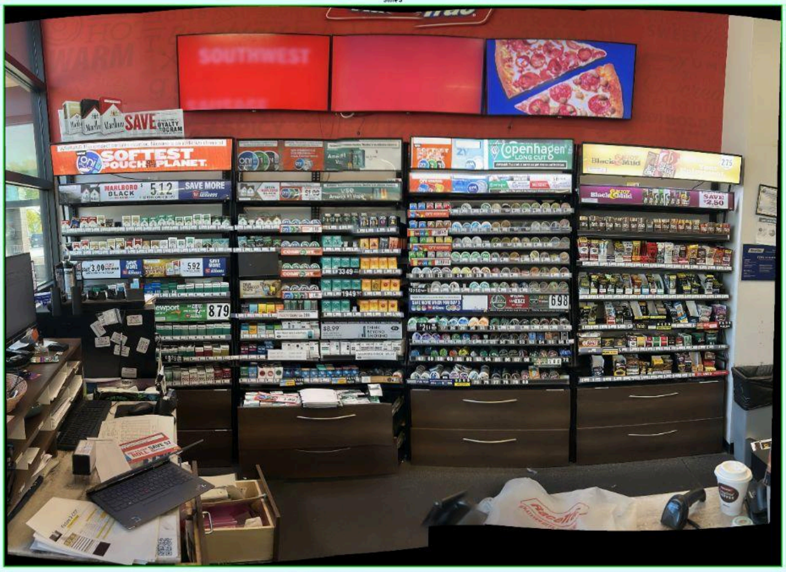
Store 1



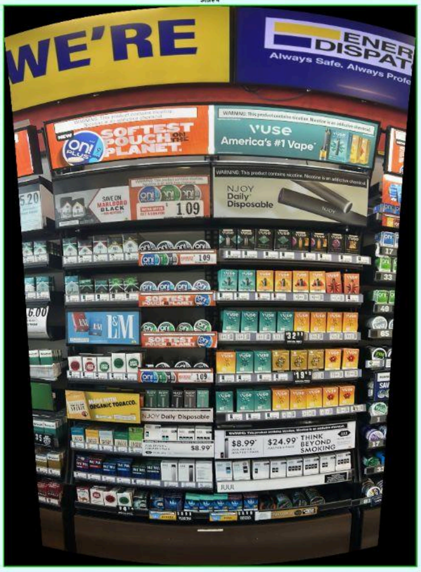
Store 2



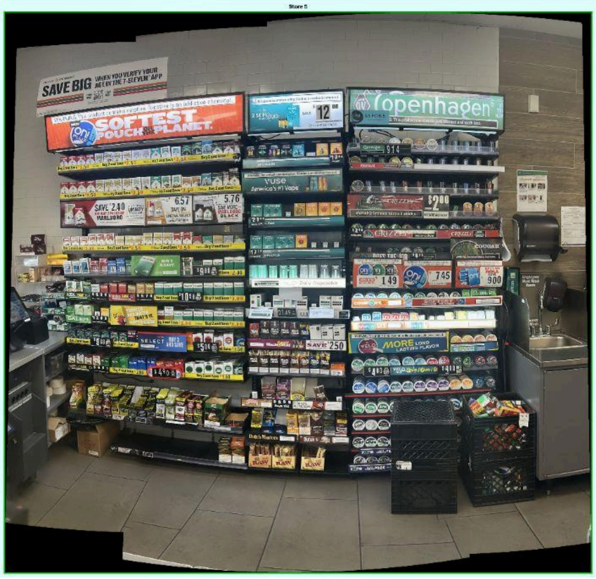
Store 3



Store 4



Store 5



Observations: cv2.Stitcher + SIFT fallback failed for **Store 1** and **Store 4** (multi-image cases) → repetitive shelf textures generate too many outlier matches → RANSAC cannot find stable homography across 3+ images.

Approach 2: SuperPoint + LightGlue

- SuperPoint → deep learned detector → robust on repetitive shelf textures
- LightGlue transformer matcher → global context matching → fewer false matches than nearest-neighbour
- RANSAC homography → tighter threshold justified by high-quality matches3.0px
- Sequential stitching left to right → alpha blend overlap → crop borders
- **Limitation**: Computationally heavy and process is slow.

Results:

Observations: For Store 1 it is complete failed because 4K resolution with repetitive shelf textures produced too few reliable matches (6 inliers from 49) → compounding bad homographies across 3+ images caused panorama to explode to 21628×12653px.
Store 4 is distorted.
Store 5 lost resolution.

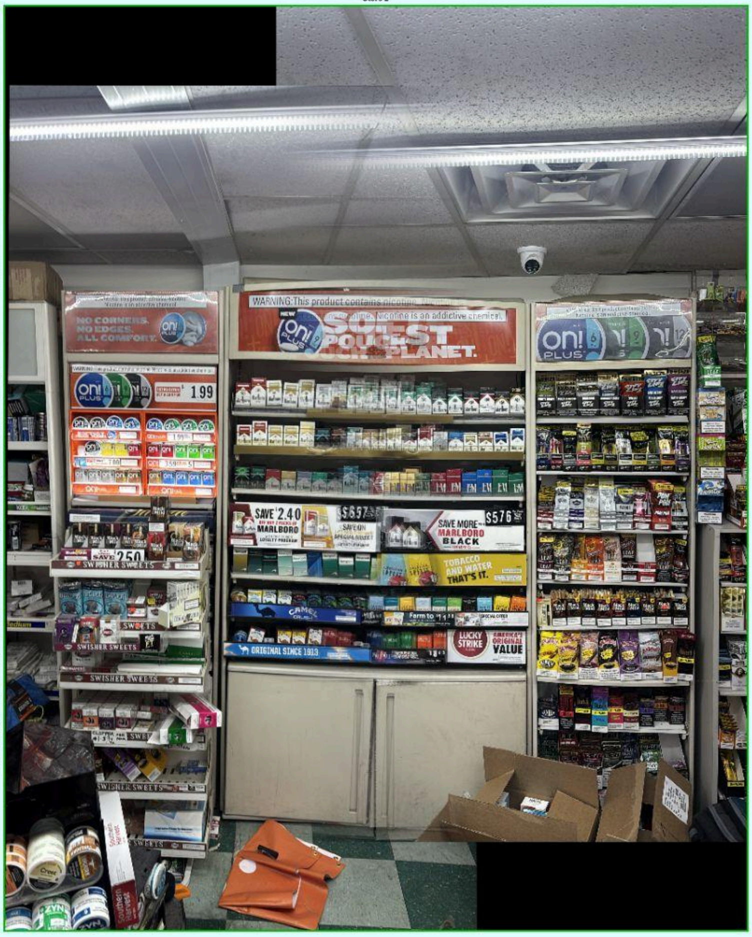
Approach 3: LoFTR

- LoFTR → closer to retail store environment `pretrained='indoor'`
- Dense matcher → operates on full image grid → no separate detect + describe → directly outputs matches end-to-end
- Resize to → fixed size required for transformer attention → scale keypoints back to original resolution for accurate warping `(480, 640)`
- Filter → removes low-quality matches in textureless shelf areas `confidence >= 0.5`
- RANSAC → high iterations handle near-duplicate points from dense matching `3.0px maxIters=10000`
- Warp + horizontal gradient alpha blend + crop borders
- Limitations: Fixed resize loses fine detail on 4K images → slow on CPU

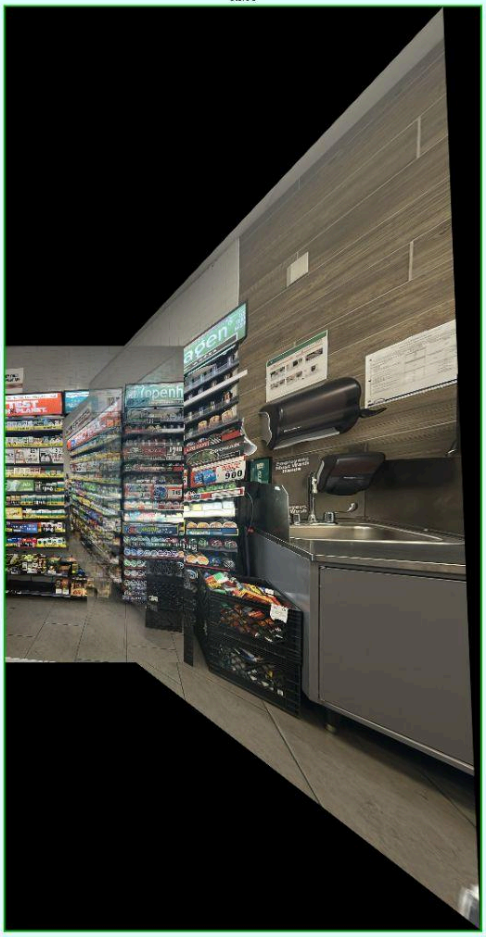
Result:

Store 1

Store 2



Store 3



Observations: Failed completely for Store 1, 3, 4, 5 → fixed resize of 4K images causes severe resolution loss → dense matching finds geometrically inconsistent correspondences despite high confidence → very low RANSAC inliers → produces heavily distorted panoramas.(480×640)

Final Approach:

RetailGlue : Semantic Product-Level Image Stitching for Retail Shelf Panoramas

(https://github.com/rempeople/RetailGlue/tree/main)

Pipeline

1. Load all store images sorted by filename → consistent left-to-right sequence
2. For each consecutive image pair → run YOLOv11 → detect product/shelf bounding boxes
3. Crop detected product regions → extract DINOv3 embeddings per crop → builds semantic identity per product
4. Use YOLO box centers as semantic anchor points + SuperPoint/LoFTR keypoints as geometric anchor points
5. Match anchor points between image pairs → filter by confidence threshold → retain only reliable matches
6. Match DINOv3 embeddings across image pairs → confirms same product is matched → prevents false matches on visually similar products
7. Compute homography matrix **H** via RANSAC → **H is the projective transform that maps pixels from source image onto destination image plane**3×3
8. Validate H → check inlier ratio → reject if too few inliers → unreliable transform
9. Project 4 corners of **source image** (image being warped) through H → compute output canvas size → apply translation to shift coordinates positive`T @ H`

10. Warp **source image** (next image in sequence) onto canvas using → aligns it with current panorama`cv2.warpPerspective`

11. Build overlap mask → use YOLO box IoU + DINOv2 similarity → detect and remove duplicate products in overlap zone

12. Apply horizontal gradient alpha blend in overlap region → smooth seam transition

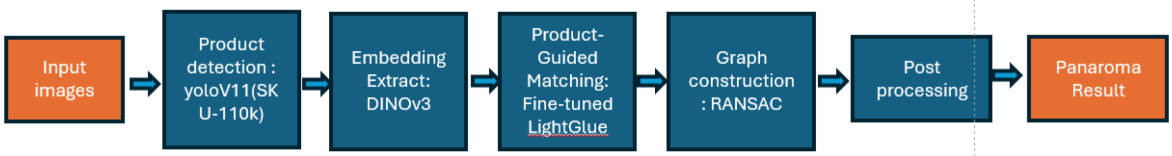
13. Crop black borders → clean canvas edges

14. Accumulate stitched result as new panorama → becomes source for next iteration

15. Repeat steps 2-14 for each next image

16. Save final panorama per store

Diagram:

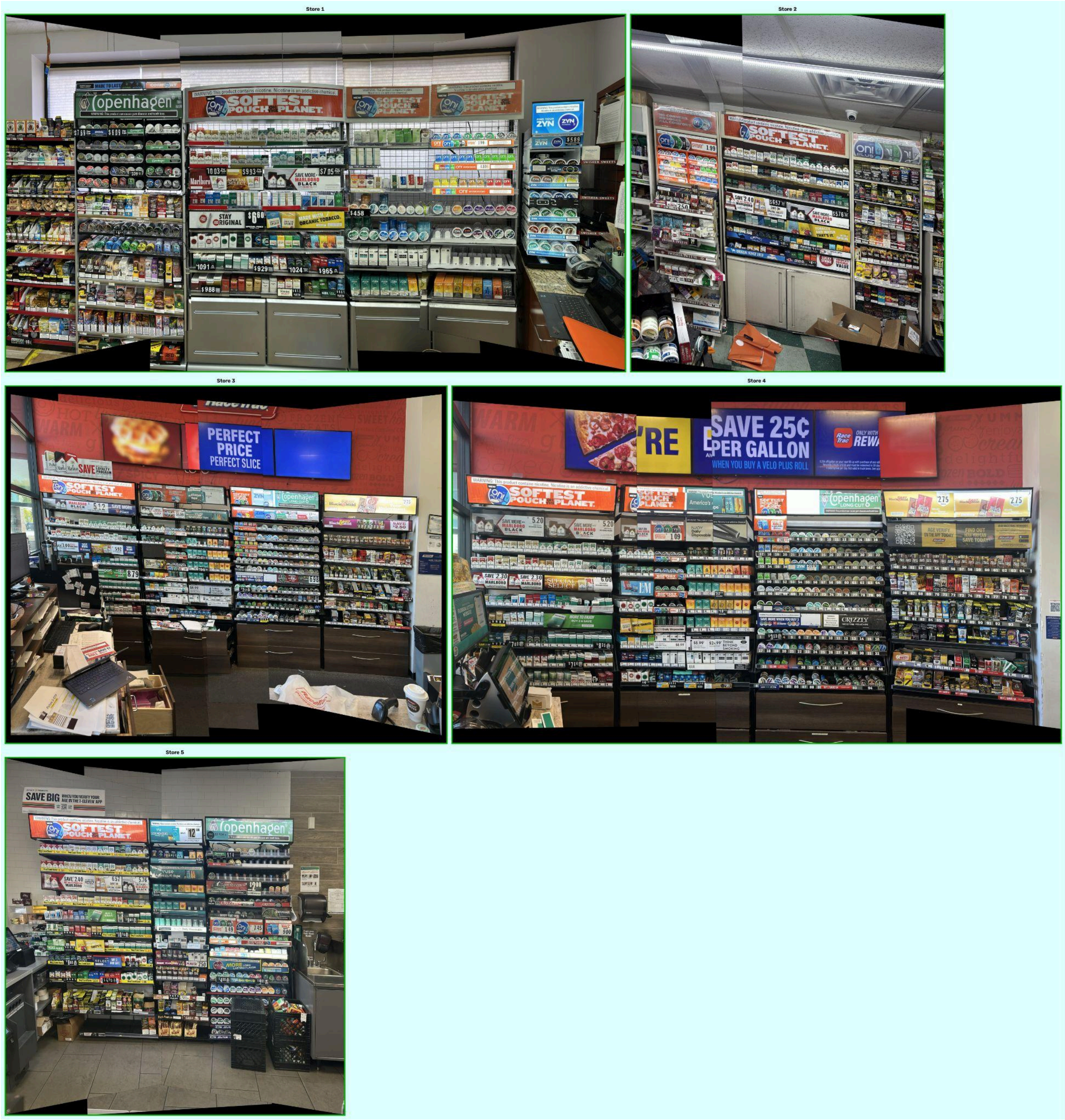


End-to-End Stitching Quality on RemRetail100_v2:

Pipeline / Matcher	Dim.	Acc.	Prec.	Rec.	F1
Local feature matching pipelines (baselines)					
SuperPoint + LightGlueStick	256	80.2	90.2	87.0	87.9
RetailGlue					
DINOv3 ViT-B/16 + LightGlue	768	80.7	89.4	87.7	88.4

RetailGlue consistently outperforms the baselines in runtime, completing medium-sized (8-image) sequences in **2.34 s** compared to **9.76 s** for DISK + LightGlue and **15.82 s** for RoMav2.

Results:



Note: Using DINOv3 ViT-S/16
Observations: RetailGlue produces panoramas that closely match the provided reference panoramas. Preserved high-resolution details and avoided duplicate product instances. The reconstructed panoramas are geometrically consistent across all evaluated stores; only Store 2 exhibits a slight tilt in the final stitched output.

Solution to fix Store 2 tilt stitched panorama:

1. RetailGlue faced Homography Estimation Issue
With only 2 images, fewer point combinations available to compute homography → USAC-MAGSAC accepts it as geometrically valid but tilted.

2. Sparse Keypoints = Less Geometric Constraint
Only product centers used → too few RANSAC inliers → small uncorrected angular error causes visible tilt.

Why cv2. Stitcher + SIFT Gives Better Result for stitching =>2 images:

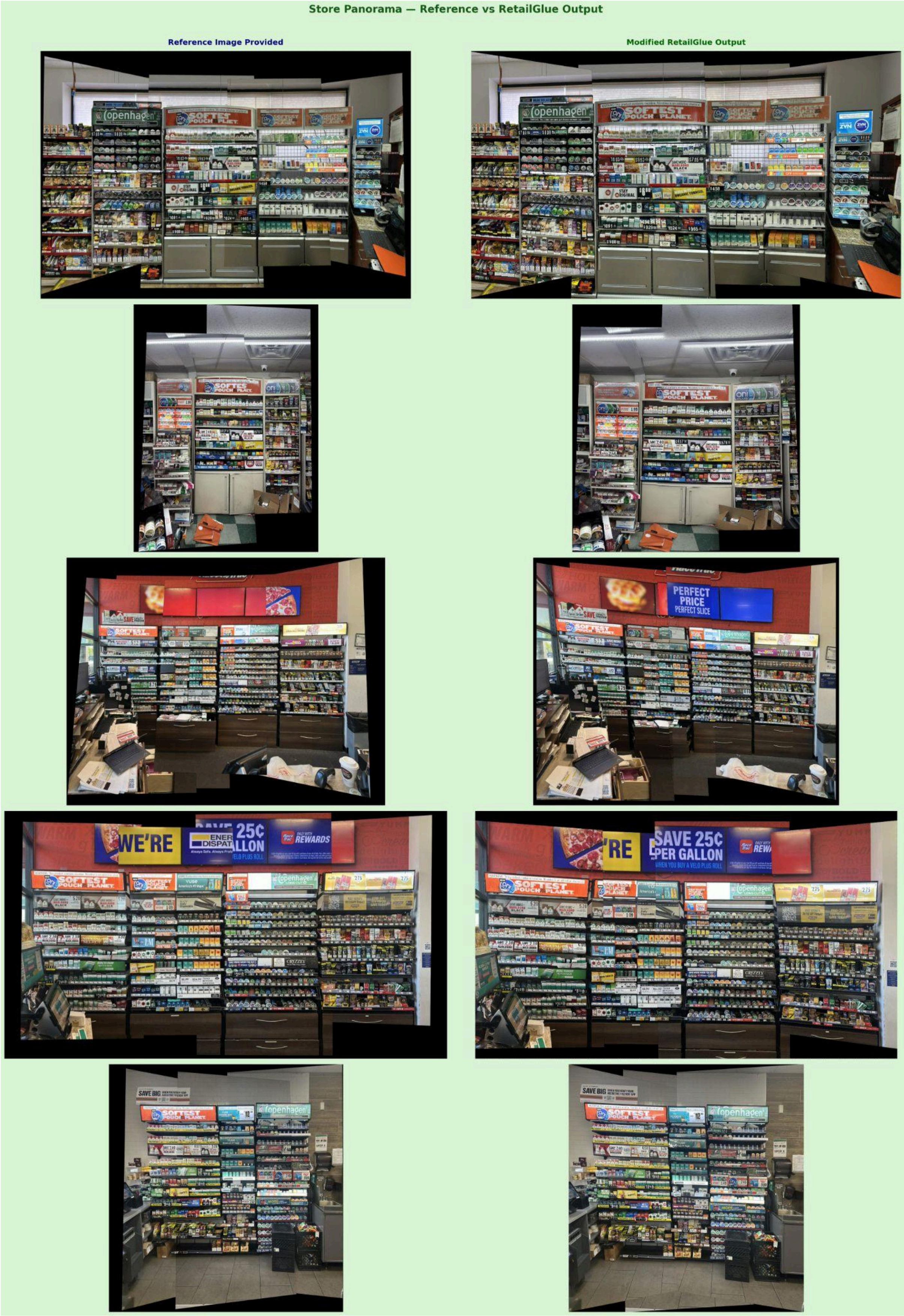
- SIFT finds **thousands of well-distributed keypoints** on 4K images
- cv2. Stitcher runs **bundle adjustment + wave correction** → tilt corrected automatically
- RetailGlue's strength is **repetitive/identical products** where SIFT gets confused by ambiguous matches

Fixed Approach added in stitcher.py for stitching 2 images:

1. **2-image pipeline bypass** — When only 2 images are detected, the existing tilt-prone graph pipeline is skipped and a dedicated fix path runs instead
2. **Hybrid stitching** — RetailGlue's product matches compute the homography; then takes over for warping, brightness correction & smooth blendingcv2.detail
3. **Tilt correction** — Homography is broken into possible rotations; the one with least tilt (closest to identity) is picked, then straightens the final panoramawaveCorrect(HORIZ)
4. **Fallback to cv2. Stitcher** — If hybrid stitching fails due to too few matches or bad homography, (SIFT-based, more keypoints) automatically takes overcv2.Stitcher
5. **Black border removal** — After warping, cleans up any black edges left around the panorama_crop_black_borders()

Results:

Comparison with provided Reference image:



Personal Contribution: Progressive RANSAC Adapted from Jia et al. (2024) into RetailGlue

From **Jia et al. 2024(** Semantic Aware Stitching for Panorama **)**(No code available), the core iterative plane-extraction logic is adapted for a RetailGlue.

RetailGlue originally used a single RANSAC to compute one global homography transform between overlapping shelf images.

Motivation for adapting Progressive RANSAC:

- A **single RANSAC** tries to fit **one global transform** to ALL correspondences across ALL depth planes
- This produces a **noisy, averaged similarity matrix** — not truly accurate for any single plane
- Result: **misalignment, ghosting, blurring** in stitched output

What **Jia et al. proposed:**

Step	Action
1	Run RANSAC with lenient threshold (0.225) → remove only gross outliers
2	Run RANSAC with strict threshold (0.2) → extract one plane's inliers → compute that plane's centroid
3	Remove remaining points spatially close to that centroid
4	Repeat Steps 2 & 3 until inlier ratio < 0.30
5	Each inlier set → compute its own similarity transformation matrix

Personal Adaptations on basis of RetailGlue's requirement:

1. **Adopted Progressive RANSAC concept** from Jia et al. 2024 — replaced RetailGlue's original single RANSAC with an iterative plane-extraction approach, so each shelf row gets its own accurate similarity transform instead of one noisy global transform
2. **Adapted plane removal criterion** — changed from Jia's centroid-proximity removal to reprojection error-based removal, using the same threshold for both RANSAC fitting and point removal to prevent drift across store deployments
3. **Added resolution-independent normalization** — coordinates normalized to a fixed range making thresholds camera-agnostic across different store hardware
4. **Introduced semantic pre-filters** — added DINOv3 embedding similarity and LightGlue confidence filters before RANSAC to reduce false matches caused by visually repetitive retail products
5. **Added plane quality scoring** — each extracted plane ranked by a DOF-adjusted score combining inlier count, embedding similarity, and reprojection error, ensuring the dominant shelf plane drives the final warp
6. **Added degeneracy check** — rejects near-collinear centroid sets common along single shelf rows where the similarity transform would be poorly constrained
7. **Integrated into RetailGlue pipeline** — added as and called from via /retailglue/stitching/progressive_ransac.pystitcher.py_calculate_homography_with_progressive_ransac()

Results:

Observations:

- Results are visually similar to single RANSAC across all stores — no regressions on Store 1, 2, 3, 4, 5
- Tested on only 5 stores — larger scale testing needed to draw strong conclusions

Progressive RANSAC Will Show Real Benefit In:

- Multi-plane shelf scenes — handles each shelf row independently instead of one dominant plane
- Varying camera angles — resolution-independent thresholds reduce need for per-store tuning

References

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2. Jia, Y., Li, Z., Zhang, L., Song, B., & Song, R. (2024). Semantic Aware Stitching for Panorama. *Sensors*, *24*(11), 3512. https://doi.org/10.3390/s24113512
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